

MAC CONTROL ELEMENT ENCODING IN 5G

Presented by : Vaishnavi , Thejas , Shruti

05-05-2026

tieto

LEARNING AND PROJECT JOURNEY

- Telecom fundamentals
- 4G to 5G evolution
- Architecture understanding
- Mentor-guided learning (courses)
- Team presentations:
 - 4G Architecture
 - VoLTE
 - 5G (NSA / SA)
 - QoS, PDU Session
- Protocol stack :
 - RRC & PDCP
 - MAC & RLC
 - PHY

Project: MAC CE Encoder

- ✓ Studied 3GPP spec
- ✓ selected 9 MAC CEs
- ✓ Incremental implementation
- ✓ Input validation & debugging
- ✓ MAC CE validation using GTEST

INTRODUCTION TO TELECOMMUNICATION



The telecommunication domain deals with transmitting information over long distances using wireless and wired technologies.



Over time, it has evolved from 1G to 5G, improving speed, latency, and overall efficiency.

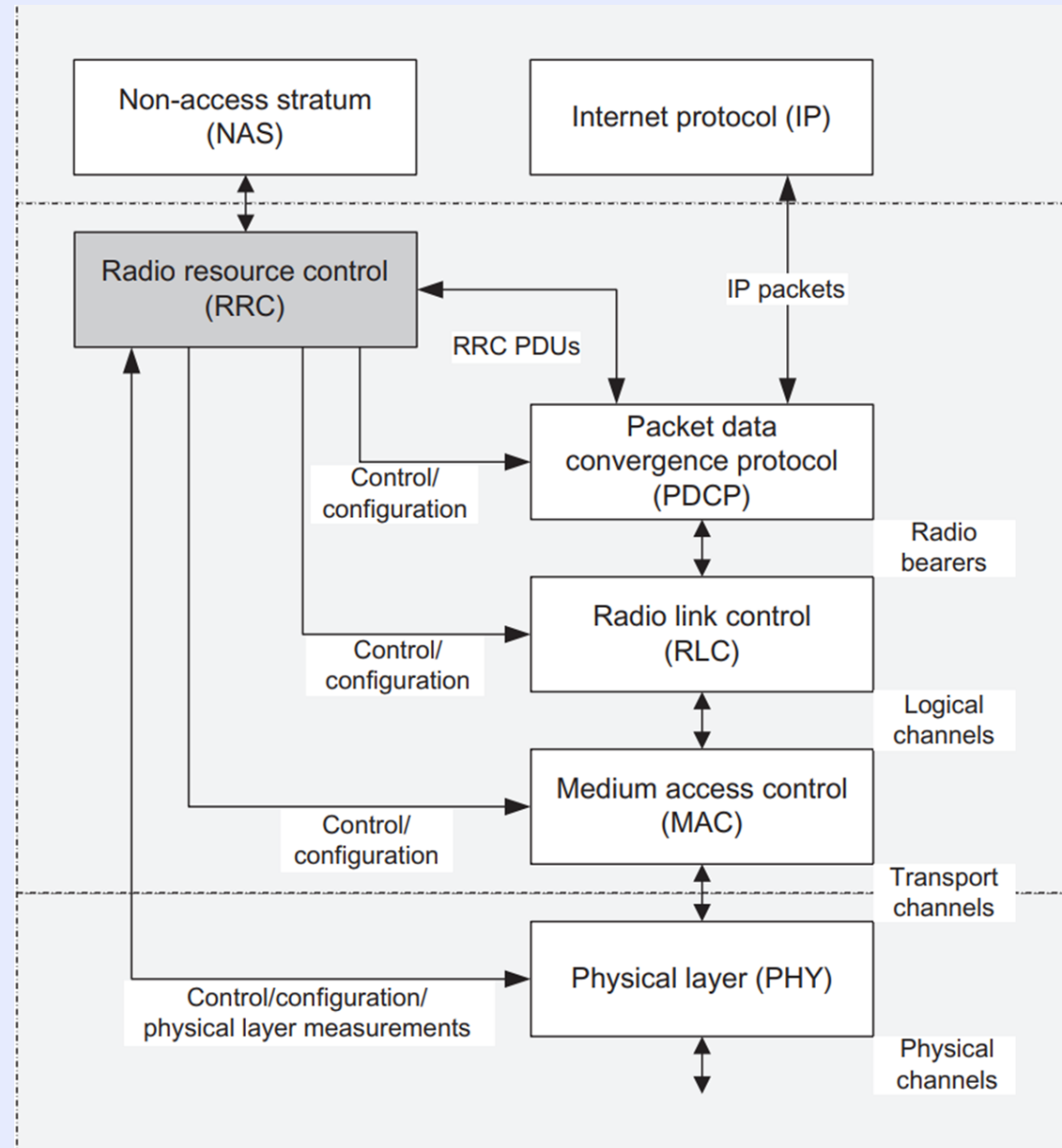


These systems follow a layered architecture, where each layer performs a specific function in communication.



Our work focuses on the MAC layer, which is responsible for handling control elements and efficient data transmission.

PROTOCOL STACK



MEDIUM ACCESS CONTROL

FUNCTIONALITIES

- Prioritization
- Scheduling
- Multiplexing
- HARQ

MAC Control element

MAC control elements are used to exchange important control information such as buffer status and power information.

These elements help the scheduler make efficient decisions for resource allocation.



OBJECTIVE



The objective of this project is to design and implement a UL MAC CE encoder that supports agreed UL MAC CEs defined in Release 18.



The encoder shall support multiple MAC CEs aggregation, enforce total MAC PDU length constraints, generate hexadecimal byte stream output, and handle invalid inputs and boundary conditions robustly.

IMPLEMENTED MAC CONTROL ELEMENTS

SHORT BSR

POWER HEADROOM REPORT(PHR)

CRNTI

RECOMMENDED BIT RATE

DSR

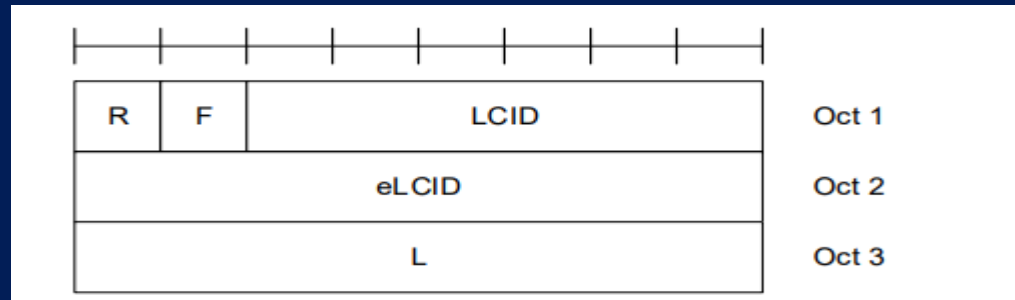
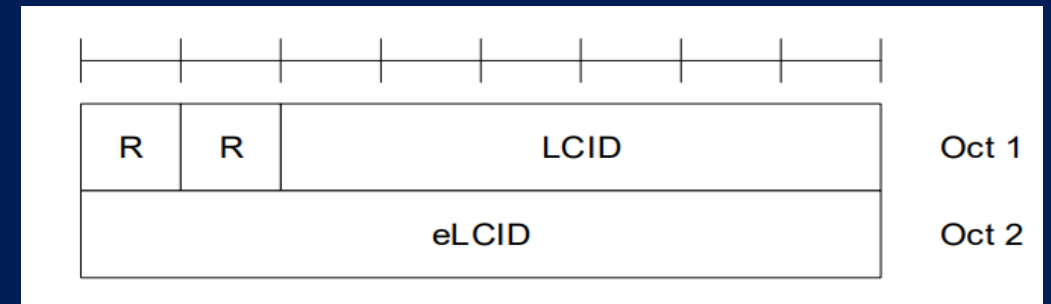
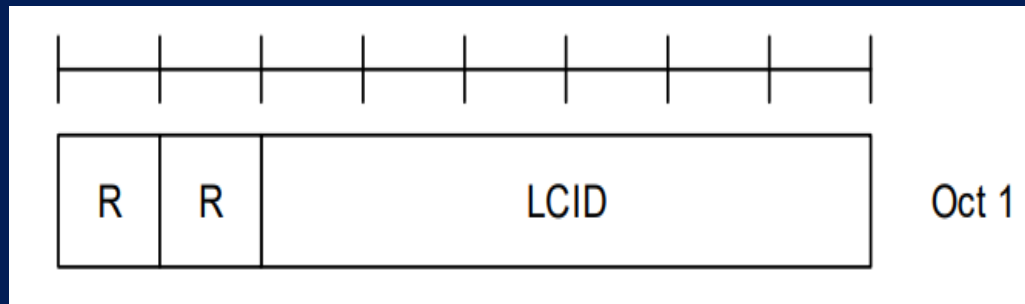
ENHANCED PHR

SL-LBT

ENHANCED BFR

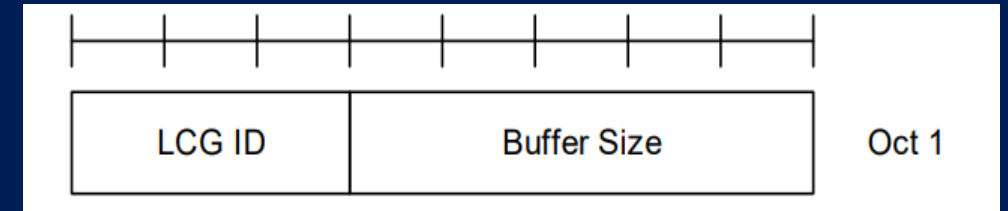
EXTENDED BSR

SUB-HEADER STRUCTURE AND FORMAT



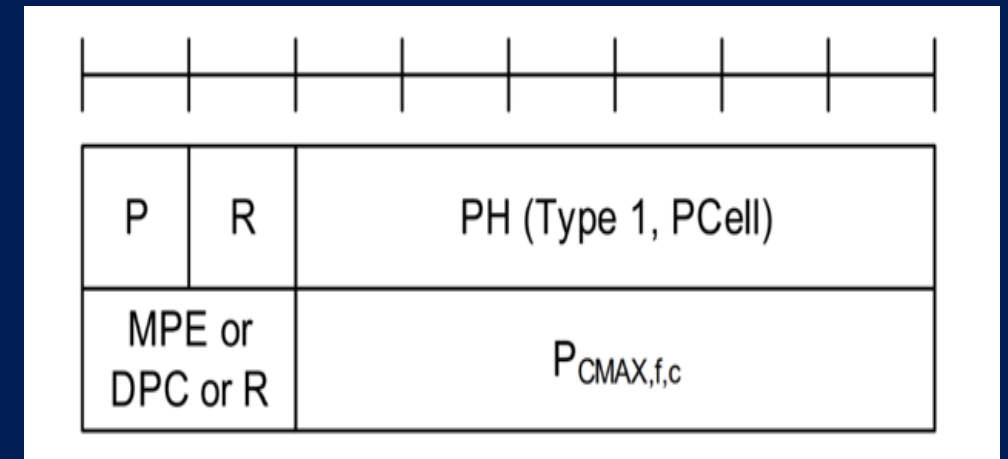
SHORT BSR MAC CE

- UE informs gNB about uplink buffer availability per Logical Channel Group (LCG).
- LCID = 61 with 1 byte subheader



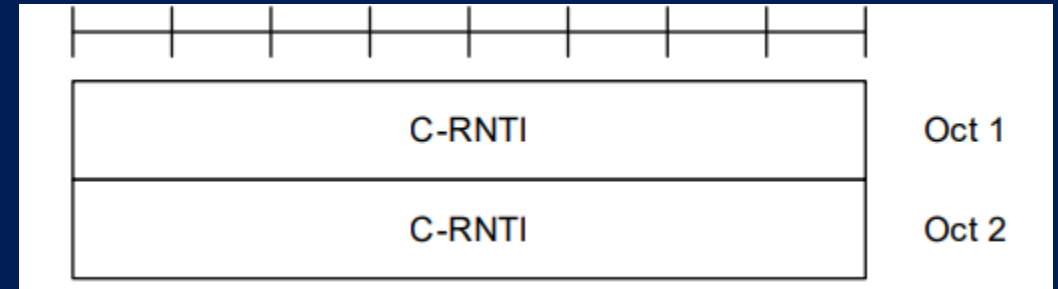
PHR MAC CE

- 2-byte payload structure: Byte 1 contains P bit, R bit, and 6-bit PH field.
- Byte 2 contains 2-bit MPE/DPC field and 6-bit PCMAX.
- LCID=57 with 1-byte subheader.



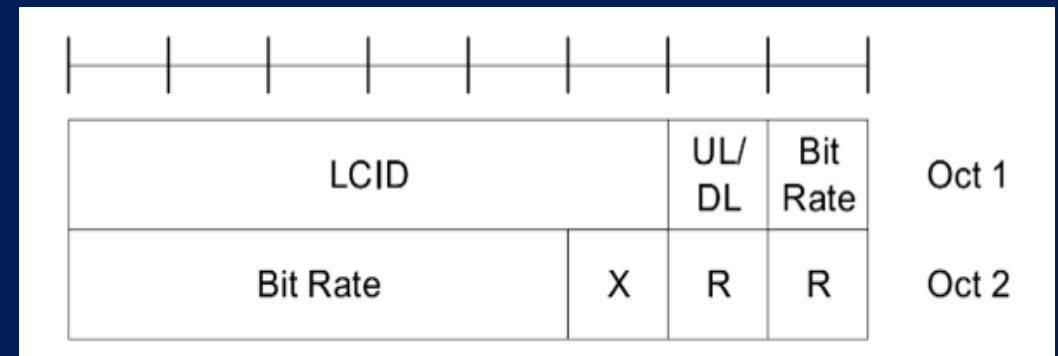
C-RNTI MAC CE

- Reports UE's 16-bit temporary identifier to the gNB during contention resolution.
- LCID=58 with 1-byte subheader.



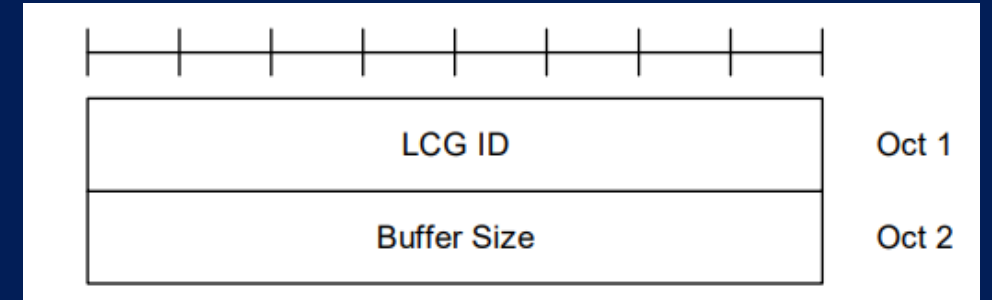
RECOMMENDED BIT RATE MAC CE

- Base station recommends transmission speed for a specific channel
- Works in both uplink and downlink direction
- LCID = 53 with 1 byte subheader
- Total size = 3 bytes



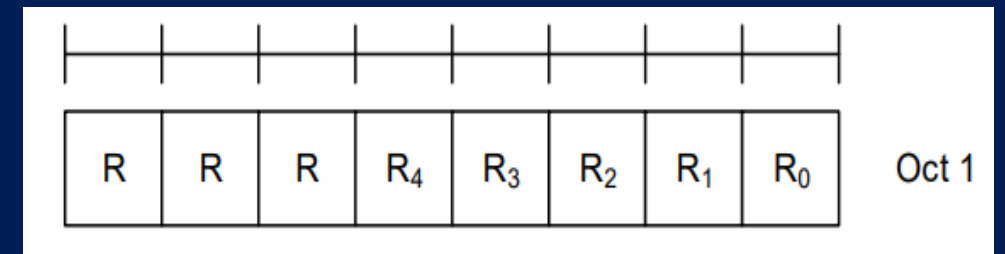
EXTENDED SHORT BSR MAC CE

- Extended BSR: groups 0–255, buffer 0–255
- LCID = 245 with 2 byte extended subheader
- Total size = 4 bytes



SL-LBT MAC CE

- Reports which sidelink channels were busy when phone tried to transmit
- Sidelink = direct phone to phone communication
- LBT = must listen before transmitting on channel
- LCID = 222 with 2 byte extended subheader
- Total size = 3 bytes



ENHANCED SINGLE ENTRY PHR FOR MULTIPLE TRP MAC CE

- Reports power headroom for two beams simultaneously
- Used when phone connects to multiple base station points (TRPs)
- LCID = 221 with 2 byte extended subheader

P	V	PH 1 (Type 1, PCell)							Oct 1
R	V	PH 2 (Type 1, PCell)							Oct 2
MPE or R		$P_{\text{CMAX},f,c}$							Oct 3

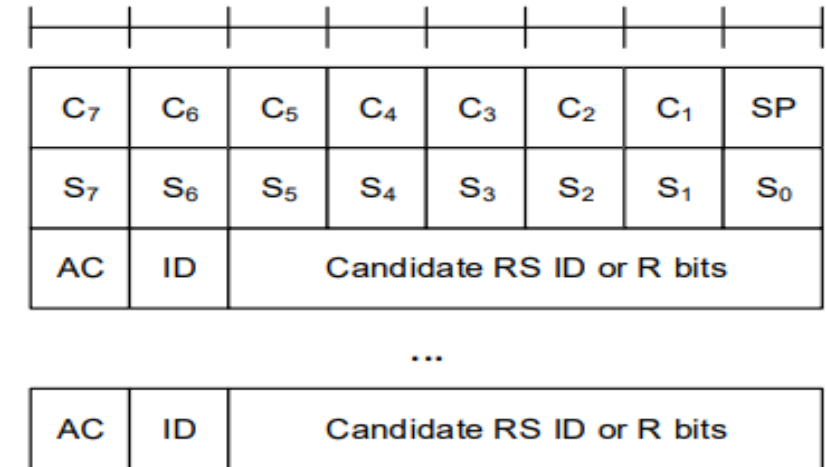
DELAY STATUS REPORT (DSR) MAC CE

- Reports delay-critical uplink data per Logical Channel Group
- Supports multiple LCGs with variable payload length
- LCID = 228 with 2 byte extended subheader

LCG ₇	LCG ₆	LCG ₅	LCG ₄	LCG ₃	LCG ₂	LCG ₁	LCG ₀	Oct 1	
BT 1	R	Remaining Time 1						Oct 2	
Buffer Size 1								Oct 3	
...									
BT m	R	Remaining Time m						Oct 2m	
Buffer Size m								Oct 2m+1	

ENHANCED BEAM FAILURE RECOVERY MAC CE

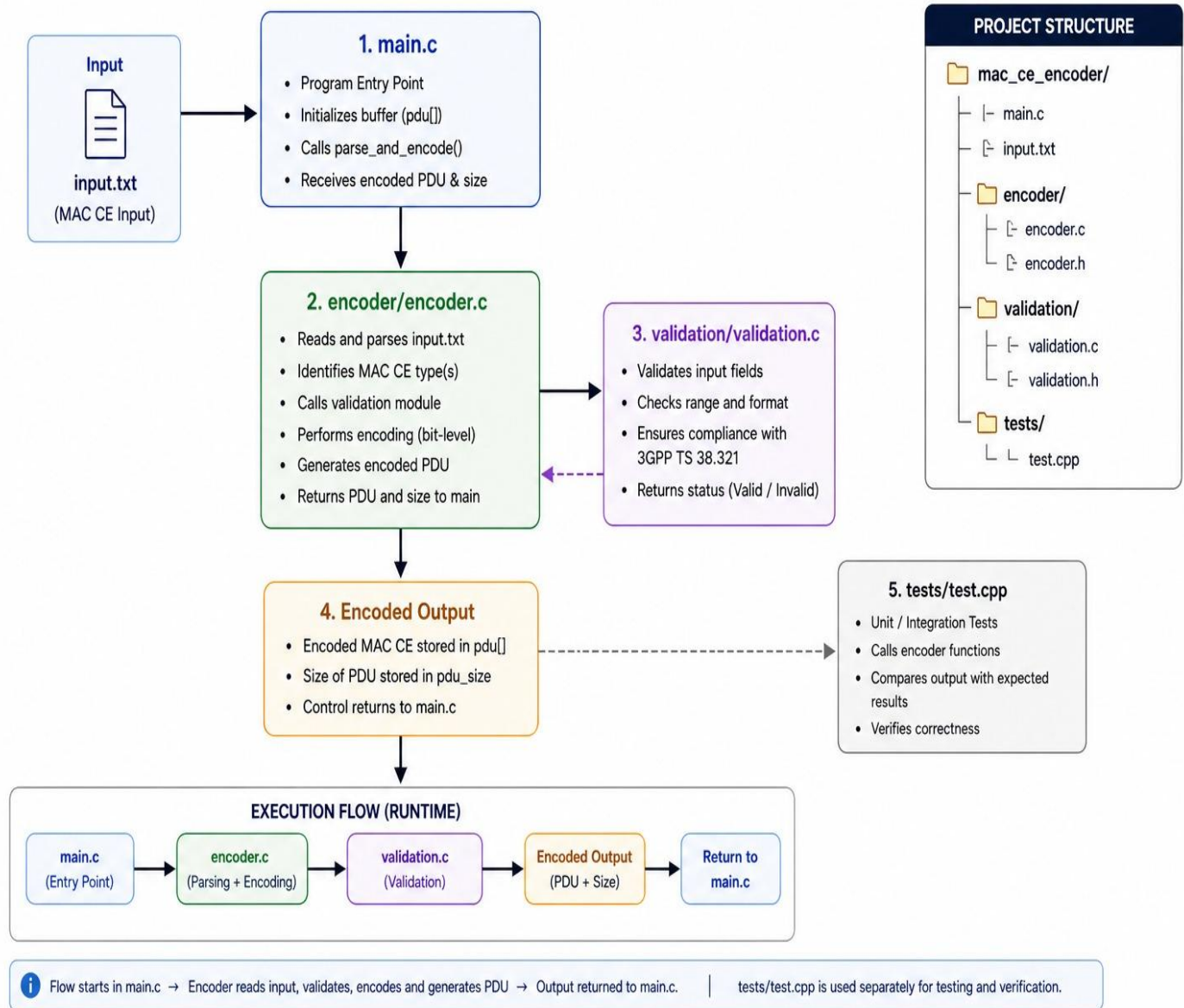
- Enables faster beam failure recovery
- Reports candidate beams to gNB
- Supports multiple beam candidates
- LCID=235 with eLCID Subheader



IMPLEMENTATION DETAILS

- Modular architecture with clear separation of Encoding, Validation, and Testing
- Input validation enforced prior to MAC CE encoding as per 3GPP TS 38.321 specifications
- Unit testing framework ensures correctness of encoded MAC PDU outputs

MAC CE ENCODER - ARCHITECTURE DIAGRAM



OUTPUT

SINGLE MAC CE

```
= encode_input.txt
1  Total pdu_size 10
2  num_ce 1
3  <sl_lbt>
4  value=12
5
6
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
MAC CE : sl_lbt
Subheader Size : 2 byte
Payload Size : 1 bytes
Total CE Size : 3 bytes (Subheader + Payload)
Encoded Bits : 00100010 11011110 00001100
Encoded Hex : 22 DE 0C
[SUCCESS] sl_lbt Encoded

===== FINAL SUMMARY =====
Total PDU Size : 10 bytes
Total Used Bytes : 3 bytes
Remaining Bytes : 7 bytes

Remaining bytes filled with 00.

Final MAC Buffer:
22 DE 0C 00 00 00 00 00 00 00
```

MULTIPLE MAC CE

```
= encode_input.txt
1  Total pdu_size 10
2  num_ce 2
3  <short_bsr>
4  lcgid=2
5  buffer=23
6
7  <dsr>
8  lcg=2
9  rt=23
10 buffer=12
11
12
13
14
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
MAC CE : short_bsr
Subheader Size : 1 byte
Payload Size : 1 bytes
Total CE Size : 2 bytes (Subheader + Payload)
Encoded Bits : 00111101 01010111
Encoded Hex : 3D 57
[SUCCESS] short_bsr Encoded

MAC CE : dsr
Subheader Size : 2 byte
Payload Size : 4 bytes
Total CE Size : 6 bytes (Subheader + Payload)
Encoded Bits : 00100010 11100100 00000011 00000100 00010111 00001100
Encoded Hex : 22 E4 03 04 17 0C
[SUCCESS] dsr Encoded

===== FINAL SUMMARY =====
Total PDU Size : 10 bytes
Total Used Bytes : 8 bytes
Remaining Bytes : 2 bytes

Remaining bytes filled with 00.

Final MAC Buffer:
3D 57 22 E4 03 04 17 0C 00 00
```

UNIT TESTING OF MAC CONTROL ELEMENT ENCODER

- Implemented using Google Test (GTEST) framework
- Validates encoding of different MAC Control Elements

Types of Test Cases Covered:

- Valid Inputs
- Missing parameters
- Extra parameters
- Invalid range
- Negative value
- Boundary conditions (Min / Max)

CHALLENGES AND RISKS

- Handling bit-level packing across byte boundaries presents significant challenges.
-

- Variable length MAC CEs like DSR and Enhanced BFR require dynamic encoding logic with strict boundary management.
-

- Ensuring proper input validation and handling invalid values and edge cases was a challenge.

KEY SOFT SKILLS DEVELOPED

Problem Solving

- Debugged encoding and test case failures
- Handled edge cases (buffer overflow, invalid inputs)

Adaptability / Learning

- Learned new tools (GTEST, encoding logic)
- Understood telecom concepts

Team Collaboration

- Discussed ideas/approach with mentor
- Followed coding standards for better readability

CONCLUSION



Successfully implemented modular MAC CE Encoder based on 3GPP TS 38.321 REL 18



Supported multiple MAC Control Elements with dedicated encoding functions



Developed scalable architecture with validation and utility modules



Verified encoded outputs using bit-level and hex-level testing



Strengthened understanding of 5G NR MAC layer and telecom protocol implementation

PROJECT RESOURCES

MAC PROTOCOL SPECIFICATION:

[3GPP TS 38.321\(R18\) – NR MAC protocol specification](#)

DOCUMENTATION: :

[MAC CONTROL ELEMENT.docx](#)

GITHUB REPOSITORY :

[shruti100/mac_ce_encoder](#)

THANK YOU

Presented by : Vaishnavi , Thejas , Shruti

05-05-2026

tieto